COVID-19 DATA ANALYTICS AND VISUALIZATION

INTRODUCTION

The COVID-19 pandemic has had a profound global impact since its outbreak in late 2019. Governments, health organizations, and researchers have relied heavily on data to monitor, understand, and respond to the crisis. In this project, we analyze COVID-19 statistics such as cases, deaths, vaccinations, and policies using structured query language (SQL) and visualize trends through Power BI dashboards. The project demonstrates how raw datasets can be transformed into meaningful insights for better decision-making.

ABSTRACT

The goal of this project is to perform end-to-end COVID-19 data analytics using a Kaggle dataset. The dataset is cleaned, normalized, and stored in SQLite for efficient querying. Analytical queries are written to answer questions such as: Which countries had the highest cases and deaths? How effective were vaccination drives? What is the impact of policies across continents? Power BI is used for visualization, presenting interactive dashboards that highlight global and country-wise trends.

This project showcases the complete data pipeline: **data cleaning → database normalization → SQL analysis → visualization**, making it a compact example of real-world data analytics.

TOOLS USED

- **Excel** – Cleaning the dataset (removing nulls, formatting, handling duplicates).
- **SQLite + DB Browser** – Database creation, normalization, and running SQL queries.
- **SQL** – Writing analytical queries and creating views.
- **Power BI** – Visualization of results (dashboards with cases, deaths, vaccinations, and policy trends).

STEPS INVOLVED IN BUILDING THE PROJECT

1. **Dataset Import & Cleaning**
    - Raw dataset imported into Excel.
    - Removed duplicates, handled missing values, reformatted dates, and removed unnecessary columns.
    - Saved both raw and cleaned datasets for reference.
2. **Database Creation & Normalization**
    - Created four normalized tables:
        - countries (population, demographics)
        - covid_cases (daily cases, deaths)
        - vaccinations (vaccination progress, boosters)
        - policies (stringency index)

- Established primary and foreign key relationships.

3. **SQL Analysis**
    - Wrote analytical queries for:
        - Top 10 countries by cases and deaths
        - Year-wise cases and deaths
        - Vaccination progress and boosters
        - Deaths per million population
        - Continent-wise COVID-19 impact
        - Vaccination vs cases correlation
    - Created **SQL views** for easy reusability and Power BI integration.
4. **Visualization in Power BI**
    - Loaded cleaned tables and SQL views into Power BI.
    - Built an interactive dashboard containing:
        - Global trends of cases and deaths
        - Vaccination progress by country
        - Scatter plot (vaccination % vs cases per million with population bubble size)
        - Continent-wise comparison charts
    - Optimized layout for professional reporting.

CONCLUSION

This project highlights the importance of structured data analytics in understanding a global crisis like COVID-19. By combining **Excel, SQL, and Power BI**, we transformed a raw dataset into actionable insights. The analysis identified the most affected countries, vaccination effectiveness, and the role of government policies.

The approach followed here is adaptable to other datasets, making it a strong example of end-to-end data analytics. In the future, this project can be extended with **predictive modeling (machine learning)** or **real-time data updates** for even deeper insights.